THE ROLE OF SMART VILLAGE INFORMATION SYSTEMS IN THE DEVELOPMENT OF THE NATIONAL ECONOMY: DIGITAL TRANSFORMATION OF THE AGRICULTURAL SECTOR

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ABSTRACT

This paper outlines a **'smart village' information system** and farm management software, adopting a comprehensive **technological and economic approach**. The envisioned technological framework includes a **React/Next.js web frontend**, a **C# Web API backend**, a **Microsoft SQL database**, and a **subsystem for scheduled background tasks and notifications**. The economic study explores the possibility of using digitalization in agriculture to improve productivity and management efficiency. In addition, foreign examples of the EU "Smart Villages" projects, Moocall, CowManager, and the World Bank "digital agriculture" plans are discussed. The paper shows that the system is a useful tool for farmers that allows them to make informed decisions using data, utilize resources efficiently, and raise productivity.

Keywords: smart village, digital agriculture, farm management, IoT, productivity.

INTRODUCTION

The global agricultural industry is confronted with challenges such as **climate change**, **resource scarcity**, **labor shortages**, and **population growth**.. The conventional form of farming is prone to ineffectual resource utilization and uncertain productivity. To solve these issues and make agriculture a sustainable, efficient, and profitable business, the deployment of digital technologies is a requirement. In this article, the necessity of a "smart village" information system in the framework of the economic development of the nation is examined and a techno-economic framework of the same is presented.

RELEVANCE OF THE PROBLEM AND THEORETICAL FOUNDATIONS

Agricultural digitalization is a top global priority. In 2017, the European Union initiated the "Smart Villages" concept in order to enhance the rural communities' socio-economic potential. There is a study in Azerbaijan that reveals the introduction of digital technologies in agriculture is at a very early stage of development and adoption of them on a wide basis could greatly enhance the performance of the sector. This need serves as the foundation for the development of this paper's technological model and economic analysis.

PROPOSED SOFTWARE MODEL

The proposed system's web frontend will be developed using the React/Next.js framework, while the server-side will be built with C# Web API services. An MS SQL Server database will be used to store large volumes of agricultural data. The architecture also includes a scheduled tasks (scheduler) component to automate tasks such as irrigation or feeding. A notification system will allow farmers to receive SMS or mobile app alerts. The system is designed to be integrated with international solutions like CowManager and Moocall to enhance its functionality.

ECONOMIC IMPACT AND BENEFITS

The adoption of digital technologies in agriculture brings significant economic benefits. These tools reduce labor and input costs, increase productivity, and improve product quality. In practice, the combined use of IoT sensors and variable-rate fertilization systems has resulted in an average productivity increase of 31% and a 23% reduction in fertilizer use. This not only optimizes costs but also minimizes negative environmental impact.

INTERNATIONAL EXPERIENCES AND LEARNING FOR THE NATIONAL ECONOMY

European Union "Smart Villages" Plan: The plan advocates for the economic and social development of rural communities through the encouragement of digital transformation. It is a key template for the development of national agricultural policies.

Moocall and CowManager Systems: These systems show the effective utilization of IoT technologies in the management of live stocks. These techniques offer effective methods that are easily customizable for the national farmers.

Digital Agriculture Roadmaps of the World Bank: These roadmaps focus on using digital technologies to increase farmers' incomes in developing countries. This approach is equally applicable to the Azerbaijani economy.

CONCLUSION AND RECOMMENDATIONS

In conclusion, a "smart village" information system is a necessary tool for farmers that helps them make decisions based on data and utilize resources more efficiently and productively. The introduction of the system will help quicken the digitization of agriculture and greatly contribute to the sustainable development of the national economy. It is advisable that the state programs be launched in order to finance such projects and the level of digital literacy among farmers be expanded. Further in the future, the incorporation of artificial intelligence in the system has the potential to offer more wide-scale forecasting and optimization capacities.

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